SakhiShield: Synopsis

Safe access. Trusted support. Greater empowerment.

Team EmpowerHer | Milestone 2 | Primary SDG: SDG 5, Gender Equality

1. Problem

Women and girls can face harassment, unsafe travel situations, or difficulty reaching immediate help during emergencies. While digital safety apps exist, weak privacy controls can expose sensitive location data through continuous tracking, and accidental triggers or false alerts can create confusion during emergencies. Low battery, poor connectivity, or language barriers can also reduce an app's usefulness when help is needed most.

2. Proposed Solution

SakhiShield is a web-based safety and empowerment platform concept where users can send SOS alerts with consent-based location sharing, a 10-second cancel option for accidental alerts, multilingual support (English and Hindi), low-data mode, and quick-access helplines (112, 1091, 181). Each SOS alert carries the user name, current location, emergency type, consent mode, and safety notes, and moves through four statuses: **Active, Acknowledged, Assisted, and Safe**. A summary page records resolved alerts and cancelled false alerts. The platform only coordinates information in a prototype setting and cannot replace human responders or police services.

3. Tools and Approach

The prototype is built with HTML (structure), CSS (design), and JavaScript (logic). It runs in a web browser and stores sample data inside the script, so no external database or server is used at this stage. Diagrams were prepared for the system architecture, SOS status workflow, data flow, and four screen wireframes. The source code is maintained in a public GitHub repository.

4. Progress Up to Milestone 2

- Send SOS form to create an alert, with a check for empty required fields and a 10-second cancel window for accidental presses
- Trusted Contacts view showing Active SOS alerts with an Acknowledge Alert button
- Status updates from Acknowledged to Assisted to Safe
- Summary page showing alerts created, acknowledged, safe, and cancelled false alerts (sample data), plus emergency helplines
- A test run of 9 steps with screenshots, design document, and README in the repository

5. Limitations and Next Steps

The current prototype does not save data after the page is refreshed, has no login, and has no real SMS or live hardware GPS connection. All figures shown are sample data created for testing, and no real impact is claimed. Planned next steps are a real database, contact verification, a campus security coordinator role, live SMS fallback integration, stronger privacy protection for contact details, and feedback from student users and campus safety representatives, to be reported only after it is actually

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